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being captured is good. AI-led soft sensors can be employed to make sure that the errors introduced by humans can be reduced to a certain extent.

Finally, looking at how process disturbances can happen on a real-time basis, it should be noted that these deviations need to be optimised for a better outcome. If optimisation is achieved, it has potential to give better outcomes than certain tools that are being deployed in the plant.

To achieve all that has been discussed above, it is necessary to bring all the data into a singular source of tools. One should not only consider new technologies for implementing these systems, but the technologies should be a combination of what is already implemented today along with application of AI models.

Most startups and firms are trying to move on with AI, ML and deep learning models only. But it is important to still continue using mathematical models of the chemical industry, physical models, thermodynamic models, and even apply in some scenarios.

Avnish Kumar, CEO, LivNSense advises on how startups should form a successful proof of concept (PoC): "Simple solution is to try to solve business problems. The idea is to try to solve problems already existing in industries. In the early days, it is helpful to start visiting plants, try to understand problems they are facing, and see how the problems can be solved by your technology and your expertise."

The physical and statistical models become the base of analytics. Then AI and ML models are applied at three stages. The first stage involves use of simple machine learning algorithms and observing the accuracy that can be achieved. Deep learning or smoothing filters, etc are applied on top of it to create a closed loop ecosystem or reinforce learning ecosystem. In implementing these algorithms, coding has to be very good for achieving high accuracy. Accuracy can vary significantly with algorithms, while affecting plant yield substantially.

Hence with all these in place a furnace can be made more intelligent, optimised, and efficiency can be improved.

Another use case is safety related

applications. It is important to leverage deep learning based algorithms with AI and vision based cameras for the safety of the workers. Safety does not end with predictive analysis, and requires human detection aspects as well. Therefore, a complete real-time monitoring of the man-machine interaction with proximity detection is critical.

The image in Fig. 3 depicts one of the incidents that was captured at one of the construction sites of the metro station, where the person was partially seen. No vehicle driver can detect the person in such situations where there is hardly the hand being captured of the person working across. To avoid such fatalities, the goal is to make machines identify such kinds of events and autonomously take actions according to safety regulations.

Conclusion

Cognitive manufacturing systems have various applications and can be applied across various kinds of equipment. By leveraging multi-source OT data for real-time early warning systems, annual savings in operational expenses of up to \$2 million can be expected. By combining real-time data with the critical process parameter data, which comes from DCS systems or historians, and building a predictive quality or predictive process optimisation, it can start delivering almost a couple of hundred thousand dollars in a few months.

Other aspects of cognitive manufacturing are in applications such as furnaces. These are the components of plants where optimisation is needed to reduce energy losses and carbon emissions. If all the learnings from thermodynamic principles are brought together to various best practices, it can really help in achieving operational cost reduction improvement by 20% and improving the run length of the furnace.

Finally, the operation aspect to it is where the instrumentation is becoming a challenge. Soft sensors can start becoming an alternative to it. Taking such areas into consideration, cognitive manufacturing can maximize profits. **EFY**

The article is based on a presentation by Avnish Kumar, CEO, LivNSense, at the IoT Startups Show 2021. Darshil Patel, who has prepared this article, is a technology journalist at EFY.